RECOVERY OF SHARED-CUBIC HIDDEN-STAR NETWORKS FROM TWO DIFFERENTIAL BOUNDARY RESPONSES

SAM VASEGHI

Abstract. Consider a network with labelled boundary terminals and unobserved interior vertices, each connected to every terminal, with no other edges in the network. On an edge incident to interior vertex c, the current law is $a_{ci}(v + \lambda_c v^3)$, where $a_{ci} > 0$ and the positive ratio λ_c is shared by all edges at that vertex. For two or more interior vertices, the differential boundary response at zero has a continuous ambiguity in the conductances. We prove that the response at zero together with the differential response at one nonzero, single-terminal bias determines all conductances and cubic ratios up to permutation of the interior vertices. The sufficient conditions are full column rank of two boundary-profile matrices and distinct component ratios in a matrix pencil. For a fixed nonzero bias, these conditions hold on an open dense set of full measure when each profile block has at least as many terminals as interior vertices. The proof combines differential-conductance reduction and established simultaneous diagonalization with an explicit recovery of physical scales. A two-center, five-terminal example illustrates every stage of the inverse. The result concerns exact differential data and the stated cubic network class; it makes no uniform stability or optimal terminal-count claim.

1. Introduction

An electrical network can contain several inaccessible junctions whose currents reach the same accessible terminals. A boundary experiment assigns terminal voltages and records the resulting terminal currents, while the interior voltages adjust to satisfy current conservation. When several hidden junctions contribute to each observed current, their separate conductances need not be determined by the aggregate linear response. In the family studied here, this failure persists through a continuous set of different positive conductances, even when the graph is known.

The network consists of r hidden centers, each joined to all m labelled terminals, with no other edges. Each center therefore forms a star, and these stars share their boundary vertices. We assume that every edge at center c has the strictly increasing current law

$$I_{ci}(v) = a_{ci}(v + \lambda_c v^3), \qquad a_{ci} > 0, \quad \lambda_c > 0. \tag{1}$$

The coefficients a_{ci} may vary independently, while the ratio of cubic to linear coefficient is common to the edges of one star. This shared ratio will allow a suitable change of boundary voltage to modify a star's response without changing its relative conductances within two chosen terminal groups.

We show that two differential boundary responses determine all the parameters in (1), subject to explicit rank and separation conditions. The first response is evaluated with every terminal at zero. For the second, a single distinguished terminal is held at a known nonzero voltage t, while the remaining terminals are held at zero. A differential response is a matrix of current derivatives with respect to terminal voltages at one operating point. Thus the data

Date: 19 October 2026.

2020 *Mathematics Subject Classification.* Primary 94C05; Secondary 15A29, 15A22.

Key words and phrases. Nonlinear resistor networks, inverse boundary problem, identifiability, differential response, simultaneous diagonalization, hidden vertices.

Wisehouse Science Communication Lab, Wisehouse AB, Ballingslöv, Sweden.

Sam Vaseghi is the corresponding author. Email: sam.vaseghi@wisehouse-publishing.com. ORCID: 0009-0006-4105-1949.

consist of two matrices, rather than two scalar current readings. At least $2r+1$ terminals suffice for our construction, and the required conditions are generic for each fixed $t \neq 0$.

The reduction from a nonlinear circuit to its differential response is established. Jekel proved that the Jacobian of the nonlinear boundary map is the response matrix of the linear network whose conductances are the derivatives of the edge laws at the equilibrium voltage drops [3]. His recovery theorem uses graphs on which the positive linear conductances can already be recovered. Nonlinear Kron reduction has also been developed in a broader network setting [4]. Our graph with $r \geq 2$ has a continuous linear ambiguity, so recovering each biased network independently does not provide the inverse studied here.

The second ingredient comes from the separation of latent components. Dreesen, Ishteva and Schoukens showed how Jacobians at several operating points lead to rectangular simultaneous diagonalization and tensor decomposition [2]. Their uniqueness criterion already covers two matrices with full-rank left and right factors and distinct component ratios. We use that established algebraic step after deriving the common factors from the hidden-voltage equations. The electrical contribution is the combination of a single-terminal bias, the resulting preservation of within-group profiles, and additional cross-responses that determine the physical conductance scales and cubic ratios.

Related nonlinear boundary recovery uses different placements of the nonlinearity. Beretta, Deng, Gandolfi and Jin study a square-lattice network with linear edge conductances and known nonlinear reactions at interior vertices [1]. Their recovery from biased differential data does not furnish the inverse below, whose nonlinear edge laws are unknown and whose interior vertices cannot be excited or measured. Our conclusion is confined to the shared-cubic hidden-star family. Within that family, it gives a complete reconstruction in a setting where zero-bias linear data cannot distinguish the conductances.

We first describe the boundary map and its linear ambiguity. The recovery theorem then follows from three calculations: the common-factor form of the biased cross-responses, separation of the hidden profiles, and recovery of their absolute scales. The proof also explains why the hypotheses are generic and which parts of the response matrices are needed.

2. Boundary response and the zero-bias ambiguity

Write $x = (x_1, \dots, x_m)^T \in \mathbb{R}^m$ for the terminal voltages and z_c for the voltage at hidden center c. Positive current on edge ci flows from terminal i to the center when $x_i - z_c > 0$. Since there are no edges between centers, each hidden voltage is determined independently by current conservation,

$$\sum_{i=1}^{m} a_{ci}\big[(x_i - z_c) + \lambda_c (x_i - z_c)^3\big] = 0. \tag{2}$$

The same equation is the stationarity condition for the function, strictly convex in z for each fixed x,

$$E_c(x,z) = \sum_{i=1}^{m} a_{ci}\left[\frac{(x_i - z)^2}{2} + \frac{\lambda_c (x_i - z)^4}{4}\right]. \tag{3}$$

Its second derivative with respect to z is at least $A_c = \sum_i a_{ci} > 0$, and the function tends to infinity as $|z| \to \infty$. Hence (2) has exactly one solution for every x. The implicit-function theorem makes that solution real analytic in the voltages and positive parameters. Minimization therefore defines a real-analytic boundary energy

$$W(x) = \sum_{c=1}^{r} \min_{z \in \mathbb{R}} E_c(x,z). \tag{4}$$

The terminal current vector and its differential response are

$$J(x) = \nabla W(x), \qquad H(x) = DJ(x) = D^2 W(x). \tag{5}$$

Differentiation of (4) gives the physical terminal currents because the derivative through the minimizing hidden voltage is multiplied by the stationarity equation (2). The matrix $H(x)$ is symmetric. Adding a common constant to all voltages changes no voltage drop, so $J(x + s\mathbf{1}) = J(x)$ and $H(x)\mathbf{1} = 0$, where $\mathbf{1}$ denotes the all-ones vector of the required size.

At zero voltage, the hidden voltages are zero and each edge has differential conductance a_{ci}. Introduce the normalized conductances

$$p_{ci} = \frac{a_{ci}}{A_c}, \qquad \sum_i p_{ci} = 1. \tag{6}$$

Eliminating a linear star's center gives the response $A_c[\mathrm{diag}(p_c) - p_c p_c^T]$, where p_c is the column vector with entries p_{ci}. Consequently, if the $m \times r$ matrix X has entries $X_{ic} = \sqrt{A_c}p_{ci}$, then

$$H(0) = \mathrm{diag}(XX^T\mathbf{1}) - XX^T. \tag{7}$$

Equation (7) displays information lost in the linear response. For an orthogonal $r \times r$ matrix O, replacing X by XO leaves XX^T unchanged. If O is sufficiently close to the identity, every entry of XO remains positive. Each positive column y of this transformed matrix defines a star with conductances

$$a_i = \left(\sum_j y_j\right) y_i. \tag{8}$$

Indeed, its total conductance is $A = (\sum_j y_j)^2$, and $\sqrt{A}\, p_i = y_i$. Thus the transformed columns give another network with exactly the same response (7). When $r \geq 2$ and X has full column rank, a small nontrivial rotation changes these stars beyond a permutation of their centers. This supplies a continuous family of distinct positive conductances with the same zero-bias data. The cubic ratios do not enter (7) at all.

The additional response will separate this family by changing the stars' relative strengths. To preserve enough common structure for comparison, we excite only one terminal and arrange the other terminals into two groups.

3. Recovery from zero and one nonzero bias

Partition all the terminals as a disjoint union

$$\{1, \ldots, m\} = B_1 \,\dot\cup\, B_2 \,\dot\cup\, \{b\}, \tag{9}$$

where b is the distinguished terminal and both B_1, B_2 are nonempty. For each center define its group masses and normalized profiles by

$$s_{cg} = \sum_{i \in B_g} p_{ci}, \qquad U_g[i, c] = \frac{p_{ci}}{s_{cg}} \quad (i \in B_g,\ g = 1, 2). \tag{10}$$

Every column of U_g is positive and has coordinate sum one. The rank condition used below says that the r profiles are linearly independent within each group. The groups may contain more than r terminals; square profile matrices are not required.

Fix $t \neq 0$, and let e_b be the coordinate vector of terminal b. At the operating point $x = te_b$, denote the hidden voltage by z_c. Equation (2) becomes

$$p_{cb}\big[(t - z_c) + \lambda_c(t - z_c)^3\big] = (1 - p_{cb})\big[z_c + \lambda_c z_c^3\big]. \tag{11}$$

Differentiating (1) at the voltage drops $-z_c$ and $t - z_c$ gives the conductance multipliers for the edges to $B_1 \cup B_2$ and to b, respectively. Define these multipliers and their weighted average by

$$u_c = 1 + 3\lambda_c z_c^2, \qquad v_c = 1 + 3\lambda_c(t - z_c)^2, \qquad D_c = (1 - p_{cb})u_c + p_{cb}v_c, \tag{12}$$

and define the bias ratio

$$\rho_c(t) = \frac{u_c^2}{D_c}. \tag{13}$$

The ratio (13) will appear as an eigenvalue of a matrix obtained from the observed responses; the separation condition in the theorem is therefore accessible from the data after the rank reduction.

Theorem 1. Consider the network (1), the partition (9), and a fixed $t \neq 0$. Suppose that U_1 and U_2 both have column rank r, and that the r numbers $\rho_c(t)$ are pairwise distinct. Then

$$H_0 = H(0), \qquad H_t = H(te_b) \tag{14}$$

determine every a_{ci} and λ_c, up to one common permutation of the hidden centers. Only the blocks H_{B_1,B_2}, $H_{B_1,b}$, and $H_{B_2,b}$ of each response matrix are needed. The number r is also determined among networks whose two profile matrices have full column rank.

For each fixed partition with $|B_1|, |B_2| \geq r$ and each fixed $t \neq 0$, the rank and separation hypotheses hold on an open dense subset of full Lebesgue measure in the positive parameter space. In particular, $m \geq 2r + 1$ is a sufficient terminal count for generic recovery by this construction.

For $r = 1$, the separation condition is empty. The theorem does not assert that $2r+1$ is the least possible number of terminals. Its uniqueness statement concerns the rank-conditioned class, a qualification whose necessity is explained in Section 7.

3.1. **The differential network and its common profiles.** We first determine how the chosen bias changes one star. For $t > 0$, the left side of (11) minus the right side is strictly decreasing as a function of z_c. It is positive at $z_c = 0$ and negative at $z_c = t$. Therefore $0 < z_c < t$. Oddness gives $t < z_c < 0$ when $t < 0$. In both cases z_c and $t - z_c$ have the sign of t, and $u_c, v_c > 1$.

At the bias, differentiation of the edge law (1) gives conductances

$$g_{ci} = a_{ci}u_c \quad (i \neq b), \qquad g_{cb} = a_{cb}v_c, \qquad G_c = \sum_i g_{ci} = A_c D_c. \tag{15}$$

Figure 1 shows this change of operating point for two centers and five terminals. The equal voltages on $B_1 \cup B_2$ give all of a center's edges to those groups the same multiplier, preserving the relative conductances that define its profiles.

To obtain the boundary response, let g_c be the column vector of these conductances. The Hessian of E_c in the boundary and hidden variables has blocks

$$\begin{pmatrix} \operatorname{diag}(g_c) & -g_c \\ -g_c^T & G_c \end{pmatrix}. \tag{16}$$

For a boundary perturbation δx, differentiated current balance gives $G_c \delta z_c = g_c^T \delta x$. Substitution into the boundary current perturbation eliminates δz_c, yielding

$$H_t = \sum_{c=1}^{r} \left[\operatorname{diag}(g_c) - \frac{g_c g_c^T}{G_c} \right]. \tag{17}$$

This is the differential-conductance reduction of [3] specialized to the present graph. It is an exact derivative identity at the prescribed bias, with no expansion in t. At zero, the same calculation uses $g_{ci} = a_{ci}$ and $G_c = A_c$, which also verifies (7).

The one-terminal bias gives (17) a common-factor structure across the two operating points. Every edge from center c to $B_1 \cup B_2$ receives the same multiplier u_c; hence its relative proportions within either group remain the columns of U_1, U_2. Since the groups are disjoint, the diagonal term of (17) does not enter their cross-block. With

$$M_0 = -(H_0)_{B_1,B_2}, \qquad M_t = -(H_t)_{B_1,B_2}, \tag{18}$$

we obtain

$$\begin{aligned} M_0 &= U_1 \operatorname{diag}(d_{0c}) U_2^T, \quad d_{0c} = A_c s_{c1} s_{c2}, \\ M_t &= U_1 \operatorname{diag}(d_{tc}) U_2^T, \quad d_{tc} = \frac{A_c u_c^2 s_{c1} s_{c2}}{D_c}. \end{aligned} \tag{19}$$

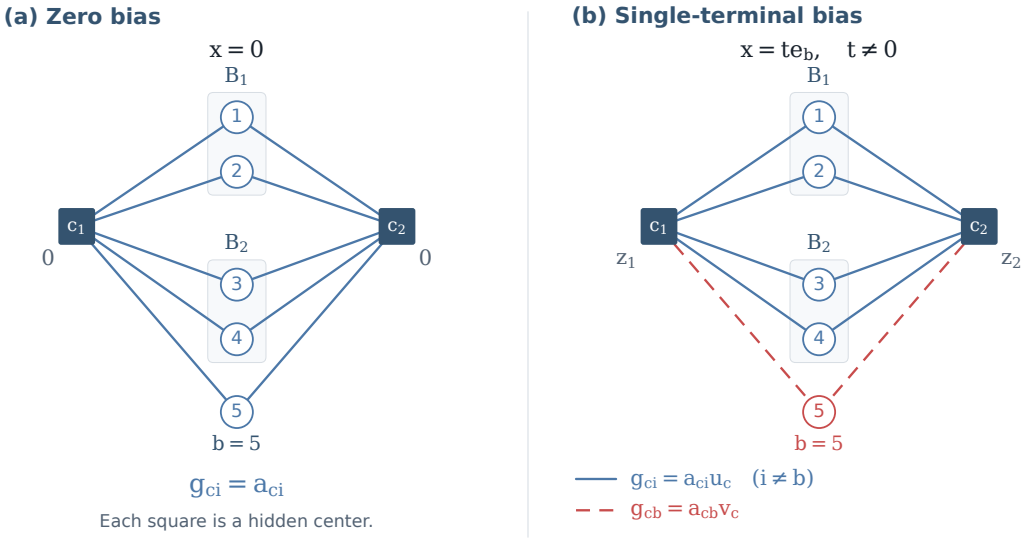


FIGURE 1. The same two-center, five-terminal graph at zero voltage and at the single-terminal bias, with $B_1 = \{1, 2\}$, $B_2 = \{3, 4\}$, and $b = 5$. At the bias, all edges from a given center to $B_1 \cup B_2$ share multiplier u_c, while its edge to b has multiplier v_c. The drawing indicates connectivity and voltage assignments, without encoding current magnitudes.

All these diagonal weights are positive, and $d_{tc}/d_{0c} = \rho_c(t)$. Thus the two matrices combine the same hidden profiles with different component weights. Their ratio information will identify the profiles that the zero-bias response left free to mix.

3.2. **Separating and normalizing the profiles.** Because both U_g have column rank r, (19) implies $\operatorname{rank} M_0 = r$. Choose matrices L, R with orthonormal columns spanning the column and row spaces of M_0, respectively, and compress the observed matrices to

$$K_0 = L^T M_0 R, \qquad K_t = L^T M_t R. \tag{20}$$

Both are $r \times r$, and K_0 is invertible. To see the information carried by their product, set $F = L^T U_1$ and $E = R^T U_2$. These matrices are invertible because L, R span the corresponding profile spaces. Equation (19) gives

$$K_t K_0^{-1} = F \operatorname{diag}(\rho_1(t), \dots, \rho_r(t)) F^{-1}. \tag{21}$$

The distinct eigenvalues determine the columns of F up to nonzero scales and permutation. Lifting these eigenvectors by L gives the columns of U_1 with the same indeterminacies. Each true profile has coordinate sum one, so division of every lifted eigenvector by its coordinate sum removes its scale and sign. That sum cannot vanish, since the lifted vector is a nonzero multiple of a positive profile.

This is the standard simultaneous-diagonalization step [2]. In tensor language, the two weight vectors give columns $(d_{0c}, d_{tc})^T$. Distinct ratios make any two such columns independent. Together with full column rank of U_1, U_2, this satisfies the two-slice uniqueness criterion for $r \geq 2$; the matrix calculation (21) also covers $r = 1$ directly.

It remains to recover U_2 in the same center order. Choose any left inverse V_1 of the recovered U_1. Then

$$V_1 M_0 = \operatorname{diag}(d_{0c}) U_2^T. \tag{22}$$

The row sums give the positive numbers d_{0c}, because the columns of U_2 sum to one. Dividing each row by its sum recovers U_2^T, without introducing a second permutation. Choose a left inverse V_2 of this recovered U_2.

The profile recovery has removed the freedom to mix centers, but it has not yet assigned an absolute conductance to any edge. Those scales are contained in the cross-responses to the distinguished terminal.

3.3. **Recovering the conductance scales.** We recover the edge conductances at both operating points by applying the following calculation to each response matrix. Let H denote H_0 or H_t, and let w_{ci} denote the corresponding edge conductance, so $w_{ci} = a_{ci}$ at zero and $w_{ci} = g_{ci}$ at the bias. Define data-derived vectors

$$\begin{aligned} d &= V_1(-H_{B_1,B_2})\mathbf{1}, \\ e &= V_1(-H_{B_1,b}), \\ f &= V_2(-H_{B_2,b}). \end{aligned} \tag{23}$$

For one center let $G_c = \sum_i w_{ci}$, and write $\sigma_{c1}, \sigma_{c2}, \sigma_{cb}$ for the fractions of that total belonging to the three groups in (9). The within-group profiles remain $U_g[:, c]$ at both operating points. The cross-blocks of (17) therefore yield, component by component,

$$d_c = G_c\sigma_{c1}\sigma_{c2}, \qquad e_c = G_c\sigma_{c1}\sigma_{cb}, \qquad f_c = G_c\sigma_{c2}\sigma_{cb}. \tag{24}$$

Set $q_{cg} = \sqrt{G_c}\,\sigma_{cg}$ for each of the three groups. Equation (24) then gives the products $d_c = q_{c1}q_{c2}$, $e_c = q_{c1}q_{cb}$, and $f_c = q_{c2}q_{cb}$. Multiplying the first two and dividing by the third isolates $q_{c1}^2 = d_c e_c / f_c$. Positivity selects its square root, after which the other two group scales follow:

$$q_{c1} = \sqrt{\frac{d_c e_c}{f_c}}, \qquad q_{c2} = \frac{d_c}{q_{c1}}, \qquad q_{cb} = \frac{e_c}{q_{c1}}. \tag{25}$$

To recover the absolute conductance scale, we use the normalization $\sigma_{c1} + \sigma_{c2} + \sigma_{cb} = 1$, which holds because the partition exhausts the terminals. Thus

$$Q_c = q_{c1} + q_{c2} + q_{cb} = \sqrt{G_c}. \tag{26}$$

The edge conductances follow as

$$w_c|_{B_g} = Q_c q_{cg} U_g[:, c] \quad (g = 1, 2), \qquad w_{cb} = Q_c q_{cb}. \tag{27}$$

Equations (23)–(27) show why the distinguished terminal need not form a third full-rank group. Once the two larger groups have separated the centers, the two cross-response vectors to b provide one coefficient per center. The three products in (24) then determine the three positive group scales. Applying the calculation to H_0 and H_t recovers all a_{ci} and g_{ci} in a common order.

3.4. **Recovering the cubic ratios and concluding uniqueness.** The recovered edge conductances give

$$u_c = \frac{g_{ci}}{a_{ci}} \quad (i \neq b), \qquad v_c = \frac{g_{cb}}{a_{cb}}. \tag{28}$$

The first ratio is independent of the chosen terminal $i \neq b$, as (15) requires. The signs established in Section 3.1 now remove the unknown hidden voltage from (12). In fact,

$$\begin{aligned} \sqrt{u_c - 1} + \sqrt{v_c - 1} &= \sqrt{3\lambda_c}(|z_c| + |t - z_c|) \\ &= \sqrt{3\lambda_c}|t|. \end{aligned} \tag{29}$$

Thus the final parameter is

$$\boxed{\lambda_c = \frac{\left(\sqrt{u_c - 1} + \sqrt{v_c - 1}\right)^2}{3t^2}.} \tag{30}$$

The hidden voltage at this bias is recovered as well,

$$z_c = t\,\frac{\sqrt{u_c - 1}}{\sqrt{u_c - 1} + \sqrt{v_c - 1}}. \tag{31}$$

Every quantity in the reconstruction is determined from the prescribed response blocks, the known bias t, and choices that affect only center order. Hence any other network in the rank-conditioned class with the same data has the same number of centers, the same normalized profiles, the same conductances at both biases, and the same cubic ratios. This proves the uniqueness assertion of Theorem 1. The remaining genericity assertion is proved next.

4. Genericity for a fixed bias

The rank and separation requirements exclude parameter coincidences rather than a compulsory failure of the model. To prove this for a fixed $t \neq 0$, use the connected parameter domain

$$\Omega = (0, \infty)^{rm} \times (0, \infty)^r \tag{32}$$

of conductances and cubic ratios. The profiles are real analytic on this domain. The hidden solution of (11), and therefore each $\rho_c(t)$, is also real analytic, since the derivative of the current-balance equation with respect to the hidden voltage never vanishes.

We exhibit one parameter choice where both profile matrices have full column rank and all ratios differ. Choose $p_{cb} = 1/2$ for every center, assign mass $1/4$ to each of B_1, B_2, and choose positive full-column-rank profiles in both groups. Such profiles exist whenever each group has at least r terminals: start from r distinct coordinate vectors, add a sufficiently small positive vector to each, and normalize their coordinate sums. Linear independence persists for sufficiently small additions.

With $p_{cb} = 1/2$, equation (11) has $z_c = t/2$. Consequently

$$u_c = v_c = D_c = 1 + \frac{3\lambda_c t^2}{4}, \qquad \rho_c(t) = 1 + \frac{3\lambda_c t^2}{4}. \tag{33}$$

Distinct positive choices of λ_c give distinct ratios. Choose one nonzero $r \times r$ minor from each profile matrix at this example. The product of these two minors and all pairwise differences $\rho_c(t) - \rho_d(t)$ is a real-analytic function on Ω which is nonzero at the example. Its zero set has empty interior and Lebesgue measure zero. Every rank failure or eigenvalue collision belongs to that set. The set where the theorem's conditions hold is also open, by continuity of the minors and ratios. It is therefore open, dense, and of full measure, completing the proof of Theorem 1.

The zero-bias ambiguity from Section 2 remains inside this identifiable class. At a parameter point satisfying the theorem with $r \geq 2$, sufficiently small orthogonal changes of X preserve positivity and full profile rank. Keeping the cubic ratios fixed, the conductances change continuously, so distinct bias ratios persist as well. These nearby networks have identical H_0, but Theorem 1 implies that their H_t cannot all coincide. The second operating point resolves an ambiguity among networks that satisfy the theorem's own assumptions.

5. An exact two-center example

Take five terminals with $B_1 = \{1, 2\}$, $B_2 = \{3, 4\}$, $b = 5$, and $t = 1$. Let

$$U_1 = \begin{pmatrix} 2/3 & 1/3 \\ 1/3 & 2/3 \end{pmatrix}, \qquad U_2 = \begin{pmatrix} 3/4 & 1/4 \\ 1/4 & 3/4 \end{pmatrix}. \tag{34}$$

Both centers have group fractions $(1/4, 1/4, 1/2)$. Choose total conductances $A_1 = 16$, $A_2 = 64$, and cubic ratios $\lambda_1 = 4/3$, $\lambda_2 = 8/3$. The edge-conductance columns are therefore

$$a_1 = \begin{pmatrix} 8/3 \\ 4/3 \\ 3 \\ 1 \\ 8 \end{pmatrix}, \qquad a_2 = \begin{pmatrix} 16/3 \\ 32/3 \\ 4 \\ 12 \\ 32 \end{pmatrix}. \tag{35}$$

Since $p_{1b} = p_{2b} = 1/2$, both hidden voltages at the bias equal $1/2$. Formula (12) gives $u_1 = v_1 = 2$ and $u_2 = v_2 = 3$. Thus the differential conductances at the second operating point are $g_1 = 2a_1$ and $g_2 = 3a_2$. The stars retain their profiles while their relative strengths change.

Their cross-block observations are

$$M_0 = \begin{pmatrix} 5/6 & 7/6 \\ 11/12 & 25/12 \end{pmatrix}, \qquad M_t = \begin{pmatrix} 2 & 10/3 \\ 5/2 & 37/6 \end{pmatrix}. \tag{36}$$

Here the cross-blocks are square and invertible, so no compression is needed. Their product is

$$M_t M_0^{-1} = \begin{pmatrix} 5/3 & 2/3 \\ -2/3 & 10/3 \end{pmatrix}. \tag{37}$$

It has eigenvalues $2, 3$, with eigenvectors proportional to $(2, 1)^T$ and $(1, 2)^T$, respectively. Normalizing their coordinate sums recovers U_1 in (34). Row normalization of $U_1^{-1} M_0$ then recovers U_2 and the weights $d_{01} = 1, d_{02} = 4$.

To recover the absolute conductance scales, we also use the cross-response vectors to the distinguished terminal:

$$\begin{aligned} -(H_0)_{B_1,b} &= (4, 6)^T, & -(H_0)_{B_2,b} &= (7/2, 13/2)^T, \\ -(H_t)_{B_1,b} &= (32/3, 52/3)^T, & -(H_t)_{B_2,b} &= (9, 19)^T. \end{aligned} \tag{38}$$

Substitution into (23) gives the coefficient triples

Operating point	Center 1: (d, e, f)	Center 2: (d, e, f)
Zero	$(1, 2, 2)$	$(4, 8, 8)$
e_b	$(2, 4, 4)$	$(12, 24, 24)$

At zero, equations (25)–(26) give group-scale triples $(1, 1, 2)$ and $(2, 2, 4)$. Their sums are 4 and 8, whose squares recover the total conductances 16 and 64. At the bias, the corresponding triples are $(\sqrt{2}, \sqrt{2}, 2\sqrt{2})$ and $(2\sqrt{3}, 2\sqrt{3}, 4\sqrt{3})$; their squared sums are 32 and 192. Formula (27) consequently returns the full columns in (35) and their respective multiples by 2 and 3.

Finally, (30) gives $4(2 - 1)/3 = 4/3$ and $4(3 - 1)/3 = 8/3$, recovering both cubic ratios. All quantities in this example are obtained from the specified response blocks. The example illustrates the complete inverse; the preceding proof supplies its validity throughout the theorem's parameter class.

6. The meaning of the differential data

Each entry $H_{ij}(x)$ describes the change in current at terminal i under a small change in voltage at terminal j, with the other boundary voltages fixed. The hidden voltages adjust during this perturbation, and their adjustment is already included in the Schur complement (17). No interior voltage or current measurement enters the reconstruction.

A voltage reference causes no additional loss of information. Since $H(x)$ is symmetric and $H(x)\mathbf{1} = 0$, a grounded response matrix determines the omitted row and column by the zero-sum identities. The vector te_b describes a voltage difference between terminal b and all the others. Grounding b instead gives the equivalent boundary vector $te_b - t\mathbf{1}$, with every other terminal at $-t$.

The selected blocks can be obtained, in the ideal derivative limit, by perturbing the terminals in $B_2 \cup \{b\}$ at each operating point and recording the required currents in $B_1 \cup B_2$. With simultaneous readout, this uses $2(|B_2|+1)$ perturbation directions across the two biases. It is a count of directions for acquiring differential responses, not a count of exact scalar measurements, and no minimality is asserted.

For a finite perturbation amplitude $h \neq 0$, analyticity gives the central-difference relation

$$\frac{J_i(x+he_j) - J_i(x-he_j)}{2h} = H_{ij}(x) + O(h^2) \tag{39}$$

as $h \to 0$. If each current reading has error bounded by η, its contribution to the error in this quotient is at most $\eta/|h|$. Smaller amplitudes therefore reduce the finite-difference truncation error while increasing sensitivity to current error. The theorem assumes the differential matrices themselves; it does not replace this estimation problem by exact recovery from finitely many nonzero-amplitude current evaluations.

Once the matrices are given, the inverse is locally smooth for fixed r on the positive parameter region with full profile rank and distinct ratios, after fixing the local ordering of the centers. The recovered scales are positive, the eigenvalues are simple, and $u_c, v_c > 1$, so the normalizations, inverses, and square roots used above are smooth locally. This gives qualitative local stability. It supplies no uniform condition number near rank failure, ratio collisions, or very small nonlinear increments.

The choice of bias also affects separation. As $t \to 0$, equation (11) gives $z_c = p_{cb}t + O(t^3)$, so (12)–(13) imply $\rho_c(t) = 1 + O(t^2)$. Thus all ratios approach one, and their differences are at most of quadratic order in t. Taking the second operating point arbitrarily close to zero can make the component-separation step poorly conditioned. The genericity statement for each fixed nonzero bias does not remove this dependence on the parameter values and chosen voltage.

7. Scope of the reconstruction

The shared cubic ratio is used at the point where a nonzero bias preserves the profiles in (19). If the ratio of cubic to linear coefficient varies independently from edge to edge within a star, the edges to $B_1 \cup B_2$ generally acquire different multipliers. The two response matrices then need not have the same profile factors, and the proof no longer applies. Unknown terminal-to-terminal edges or hidden-to-hidden edges likewise change the cross-response formula. The theorem concerns the specified graph and constitutive family, including the absence of these additional edges.

The cubic law must also hold exactly over the voltage range used. If an edge law is $a_{ci}(v + \lambda_c v^3) + O(v^5)$ near zero, unknown higher-order terms influence a finite-bias derivative. In that situation, (15) and the square-root recovery (30) cannot be applied unchanged. A theorem based on low-order derivatives at zero would be a different data statement and would require its own assumptions and reconstruction.

The qualification on the number of hidden centers is structural. A single star with conductances a_{ci} and ratio λ_c can be replaced by two stars with conductances θa_{ci} and $(1-\theta)a_{ci}$, both with the same ratio, for $0 < \theta < 1$. Their stationarity equations have the same hidden solution as the original star, and their boundary currents sum to its current for every x. Consequently even the entire nonlinear boundary map cannot distinguish these representations. The duplicate profiles violate the full-column-rank hypothesis. Within the class where both profile matrices have full column rank, $r = \operatorname{rank} M_0$ is determined; outside that class an unrestricted hidden-vertex count is not.

The contribution is an explicit inverse for this family of nonlinear networks. Differential-conductance reduction [3,4] gives two linear response representations, and simultaneous diagonalization [2] separates their common profiles. The single-terminal bias and the scale formulas (23)–(30) connect those representations to the original edge parameters. For $r \geq 2$, the result recovers conductances that zero-bias linear observations leave continuously

ambiguous, while stating the graph, coefficient, rank, and data restrictions under which that recovery holds.

Use of Generative-AI tools declaration

AI tools assisted the formulation, literature comparison, proof development, and preparation of this manuscript within the Mathematics Knowledge and Connections Program.

The tools used were OpenAI ChatGPT and Codex. Their assistance concerned the reconstruction and genericity arguments in Sections 2–4, the exact example in Section 5, the literature comparisons in Sections 1 and 7, and drafting and revision throughout the manuscript. The author is responsible for the mathematical claims, references, and final text.

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Wisehouse Science Communication Lab, Wisehouse AB, Ballingslöv, Sweden
Email address: sam.vaseghi@wisehouse-publishing.com
URL: https://orcid.org/0009-0006-4105-1949